AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	HCA Information Technology & Services Data Center, TN -- station KBNA, America/Chicago
Building	OSM 819981888 OSM way 819981888
Facade gap	not applicable -- one building, no second facade
Weather record	43,795 real hourly records from KBNA
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-01-01 (recirc_edge)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 10 of 24 hours with 2 mode changes. 7 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	10 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	10 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) 7 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	7.220	21.0	7.220	0.856
01	FREE	yes	-	7.500	21.0	7.780	0.730
02	FREE	yes	-	7.500	21.0	7.780	0.650
03	FREE	yes	-	8.330	21.0	8.890	0.826

04	FREE	yes	-	8.890	21.0	9.440	0.767
05	FREE	yes	-	9.170	21.0	10.000	0.661
06	FREE	yes	-	10.285	21.0	10.560	0.655
07	FREE	yes	-	13.885	21.0	14.440	1.222
08	FREE	yes	-	16.945	21.0	17.220	1.682
09	FREE	yes	-	19.725	21.0	20.000	2.114
10	mechanical	no	dew point	21.945	21.0	20.560	2.125
11	mechanical	no	dry-bulb	23.610	21.0	22.780	1.630
12	mechanical	no	dry-bulb	24.445	21.0	23.330	1.229
13	mechanical	no	dry-bulb	24.445	21.0	23.890	1.092
14	mechanical	no	dry-bulb	24.720	21.0	23.330	0.947
15	mechanical	no	dry-bulb	23.615	21.0	21.670	0.858
16	mechanical	no	dry-bulb	22.495	21.0	19.440	1.104
17	mechanical	yes	switch budget	19.995	21.0	16.110	1.089
18	mechanical	yes	switch budget	16.670	21.0	12.220	0.988
19	mechanical	yes	switch budget	14.725	21.0	10.560	1.254
20	mechanical	yes	switch budget	13.060	21.0	10.560	1.272
21	mechanical	yes	switch budget	10.835	21.0	10.000	1.101
22	mechanical	yes	switch budget	10.000	21.0	9.440	1.151
23	mechanical	yes	switch budget	9.725	21.0	8.890	0.959

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.220 C, which is 13.780 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.856 C, and the margin is measured, not chosen: 0.856 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.500 C, which is 13.500 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.730 C, and the margin is measured, not chosen: 0.730 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.500 C, which is 13.500 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.650 C, and the margin is measured, not chosen: 0.650 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.330 C, which is 12.670 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.826 C, and the margin is measured, not chosen: 0.826 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.890 C, which is 12.110 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.767 C, and the margin is measured, not chosen: 0.767 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.170 C, which is 11.830 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.661 C, and the margin is measured, not chosen: 0.661 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.285 C, which is 10.715 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.655 C, and the margin is measured, not chosen: 0.655 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.885 C, which is 7.115 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.222 C, and the margin is measured, not chosen: 1.222 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.945 C, which is 4.055 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.682 C, and the margin is measured, not chosen: 1.682 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 19.725 C, which is 1.275 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 2.114 C, and the margin is measured, not chosen: 2.114 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

10:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 21.945 C would have passed. The outside air is too HUMID: the dew-point bound is 16.11 C against a 15.0 C maximum, failing by 1.11 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.610 C against a 21.0 C limit, so it fails by 2.610 C. It would take a limit 2.610 C higher, or a bound 2.610 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.445 C against a 21.0 C limit, so it fails by 3.445 C. It would take a limit 3.445 C higher, or a bound 3.445 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.445 C against a 21.0 C limit, so it fails by 3.445 C. It would take a limit 3.445 C higher, or a bound 3.445 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 24.720 C against a 21.0 C limit, so it fails by 3.720 C. It would take a limit 3.720 C higher, or a bound 3.720 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 23.615 C against a 21.0 C limit, so it fails by 2.615 C. It would take a limit 2.615 C higher, or a bound 2.615 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 22.495 C against a 21.0 C limit, so it fails by 1.495 C. It would take a limit 1.495 C higher, or a bound 1.495 C tighter, to change this hour.

17:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.995 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.670 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.725 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.060 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.835 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.000 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.725 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes

per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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